Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/30/2006
PI Name:	Wiseman, Robert W Ph.D.		
Project Title:	Calcium Homeostasis and muscle phenotype: Role of cellular Energetics		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Muscle Alterations and Atrophy Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	48824	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	07/01/2001	**End Date:**	10/31/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):			
Grant/Contract No.:	MA.002.10		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Exposure of skeletal muscle to space flight results in a significant loss of mass and a shift in the phenotype from slow to fast muscle isoforms. To a limited extent, astronauts are able to ameliorate this remodeling of muscle tissue through exercise. If the mechanistic link between physiologic function and phenotype were better understood, design of countermeasures using combinations of exercise protocols and pharmaceuticals could be employed to increase the efficacy of training while on space missions. We propose that altered physiologic function signals the initiation of the remodeling process through Ca+2 sensitive transcription factors (CSTFs) which are activated through changes in two homeostatic processes; mitochondrial ATP synthesis and sarcoplasmic reticulum (SR) ATPase Ca+2 handling. It is our assertion that alterations in phenotype in response to changes in load bearing or any other metabolic stress involves processing information from the physiology in the form of feedback from these two homeostatic processes. We use an integrative approach to study this problem in isolated superfused skeletal muscles using a combination of non-invasive techniques (31P-NMR and fluorescence spectroscopy and mechanics) and molecular techniques. In the first Aim we		

	determine the sensitivity of cytosolic Ca+2 handling to metabolic loads induced by electrical pacing and metabolic inhibitors. In the second aim we test the response of CSTFs to alterations in Ca+2 homeostasis using ionophores, SR ATPase inhibitors as well as the metabolic stresses we develop in Aim 1. We believe once the mechanistic link is established that we may be able to design countermeasures to mask the loss of mechanical loading by direct manipulation of cytosolic Ca+2 and more effectively stave off the changes occurring in limb musculature.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding the basic mechanisms involved in muscle remodeling and/or breakdown will have a dramatic impact on all types of muscle wasting conditions experienced on Earth as well as direct impact on the role of muscle activation in maintaining health and well-being for people of all ages
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)